

Lihu Xu

Applied Probability, Statistics, and Machine Learning

University of Macau, E11, Avenida da Universidade, Taipa, Macau, China

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Google Scholar

Education

10/2004-01/2008	Ph.D.	Imperial College London, UK
09/2001-06/2004	M.S.	Peking University, Beijing, China
09/1997-07/2001	B.S.	Shandong University, Jinan, China

Academic Appointments

08/2026–	Professor,	University of Macau, Macau
01/2026–08/2026	Associate Professor (tenured),	Michigan State University, East Lansing, US
08/2019–08/2026	Associate Professor,	University of Macau, Macau
01/2024-07/2024	Visiting Faculty,	Harvard University, Cambridge, US
01/2014-07/2019	Assistant Professor,	University of Macau, Macau
09/2013-12/2013	Visiting Scholar,	Boston College, US
09/2011-11/2013	Lecturer (permanent),	Brunel University London, UK
10/2010-09/2011	Postdoc,	Technische Universität Berlin, Germany
10/2008-08/2010	Postdoc,	Eindhoven University of Technology, Netherlands
09/2007-08/2008	Visiting PhD & Postdoc,	Bonn University, Germany

Research

• Research Interests

• Applied Probability

Stein's method, Malliavin calculus, Convergence of stochastic algorithms in machine learning (e.g. SGD, SVRG), Ergodic theory of SPDEs, Large deviations, Rare events of SDEs and SPDEs, Numerical methods for SDEs

• Statistics and Machine Learning

High dimensional robust estimation, High dimensional time series, High dimensional distribution estimation via DNN, Langevin Monte Carlo sampling, Hamilton dynamics sampling, Heavy tailed distribution sampling

• Publications ¹

• Submitted Journal Papers (6)+Conference Papers (1)

- Guo, Yaoyuan; Mo, Yingjun; **Xu, Lihu**; Kong, Fang
Contextual linear bandits for matching markets
submitted to **NeurIPS 2026**.
- Li, Xiang; Liu, S. Jun; Sun, Qiang; **Xu, Lihu**
Tuning free Catoni type joint robust estimation,
submitted to **Journal of Machine Learning Research**, arXiv:2511.11054.
- Pei, Qiyang; **Xu, Lihu**
Functional central limit theorem for Euler–Maruyama scheme with decreasing step sizes,
submitted to **Bernoulli**, arXiv:2511.04019.
- Cai, Zhijun; Li, Xiang; **Xu, Lihu**
Berry-Esseen bounds and moderate deviations for Catoni-type robust estimation
submitted to **Electronic Journal of Statistics**, arXiv:2602.12589.
- Jin, Xinghu; Wang, Yu; Pang, Guodong; **Xu, Lihu**
Approximation of the steady state for piecewise stable Ornstein-Uhlenbeck processes arising in queueing networks,
submitted to **Journal of Applied Probability**, arXiv:2405.18851.
- Wang, Yu; Li, Guodong; Xiao, Zhijie; **Xu, Lihu**; Zhang, Wenyang
Robust estimation for high-dimensional time series with heavy tails,
submitted to **Statistica Sinica**, arXiv:2411.05217.
- Deng, Changsong; Wang, Lin; **Xu, Lihu**
Quantitative bounds for high dimensional entropic CLT,
submitted to **Transactions of the American Mathematical Society**, arXiv:2604.05861.

• Published (Accepted) Journal Papers (64)+Conference Papers (1)

- Lu, Jianya; Pan, Bo; Wang, Yafei; **Xu, Lihu**; Jiang, Bei; Kong, Linglong
Robust empirical risk minimization from a single dependent trajectory: non-asymptotic theory under infinite variance,
Statistics and Trustworthy AI for Cross (X)-Domain Acceleration Conference 2026 (STAI-X 2026), Harvard

¹Underlined authors are Ph.D. students, visiting Ph.D. students, former Ph.D. students, or postdoctoral researchers mentored by Lihu Xu.

University.

2. Guillin, Arnaud; Wang, Yu; **Xu, Lihu**; Yang, Haoran
Error estimates between SGD with momentum and underdamped Langevin diffusion,
Annals of Applied Probability (accepted).
3. Chen, Peng; **Xu, Lihu**; Zhang, Xiaolong; Zhang, Xicheng
 W_d convergence rate of EM schemes for invariant measures of supercritical stable SDEs,
accepted for publication in **SIAM Journal on Mathematical Analysis**.
4. Deng, Changsong; Li, Xiang; Schilling, Rene; **Xu, Lihu**
Total variation distance between SDEs with stable noise and Brownian motion,
accepted for publication in **Journal of Applied Probability**.
5. Chen, Peng; Xiong, Jie; **Xu, Lihu**; Zheng, Jiayu
Quantitative diffusion approximation for the neutral-alleles Wright-Fisher Model with mutations,
Potential Analysis, Vol. 64 (4), Article Number 56, 2026.
6. Wang, Yu; Xiao, Yimin; **Xu, Lihu**
Phase transition in the EM scheme of an SDE driven by α -stable noises with $\alpha \in (0, 2]$,
Journal of Computational and Applied Mathematics, Vol. 474, 116938, 2026.
7. Xie, Fang; **Xu, Lihu**; Yao, Qiuran; Zhang, Huiming
Distribution Estimation of Contaminated Data via DNN-based MoM-GANs,
Journal of Multivariate Analysis, Vol. 212, 105571, 2026.
8. Li, Xiang; Wang, Feng-Yu; **Xu, Lihu**
Unadjusted Langevin algorithm for SDEs with Hölder drift,
Science China Mathematics, Vol. 68, 3017–3042, 2025.
9. Lu, Jianya; Mo, Yingjun; Xiao, Zhijie; **Xu, Lihu**; Yao, Qiuran
Distribution estimation and change-point detection for time series via DNN-based GANs,
Machine Learning, Vol. 114, No. 208, 2025.
10. Deng, Changsong; Schilling, Rene; **Xu, Lihu**
Optimal Wasserstein-1 distance between SDEs driven by Brownian motion and stable processes,
Bernoulli, 31 (3), 1834–1857, 2025.
11. Chen, Peng; Jin, Xinghu; Xiao, Yimin; **Xu, Lihu**
Approximation of the invariant measure for stable SDE by the Euler-Maruyama scheme with decreasing step-size,
Advances in Applied Probability, 1–31, 2025.
12. Jin, Xinghu; Pang, Guodong; **Xu, Lihu**; Xu, Xin
An approximation to the invariant measure of the limiting diffusion of $G/Ph/n+GI$ queues in the Halfin-Whitt regime and related asymptotics,
Mathematics of Operations Research, no. 2, 783–812, 2025.
13. Li, Xiang; **Xu, Lihu**; Yang, Haoran
Stable central limit theorem in total variation distance,
Journal of Theoretical Probability, 38(1), 16, 1–51, 2025.
14. Jiang, Hui; **Xu, Lihu**; Yang, Qingshan
Functional large deviations for Stroock’s approximation to a class of Gaussian processes with application to small noise diffusions,
Journal of Theoretical Probability, 37, no. 4, 3015–3054, 2024.
15. Fan, Xiequan; Hu, Haijuan; **Xu, Lihu**
Normalized and self-normalized Cramér-type moderate deviations for Euler–Maruyama scheme for SDE,
Science China Mathematics, no. 8, 1865–1880, 2024.
16. Chen, Peng; Nourdin, Ivan; **Xu, Lihu**; Yang, Xiaochuan
Multivariate stable approximation in Wasserstein distance by Stein’s method,
Journal of Theoretical Probability, 37(1), 446–488, 2024.
17. Chen, Peng; Deng, Changsong; Schilling, Rene; **Xu, Lihu**
Approximation of the invariant measure of stable SDEs by an Euler–Maruyama scheme,
Stochastic Processes and Their Applications, 163, 136–167, 2023.
18. **Xu, Lihu**; Yao, Fang; Yao, Qiuran; Zhang, Huiming
Non-asymptotic guarantees for robust statistical learning under infinite second moment assumption,
Journal of Machine Learning Research, 24, 1–46, 2023.
19. Nersesyan, Vahagn; Peng, Xuhui; **Xu, Lihu**
Large deviations principle via Malliavin calculus for the Navier-Stokes system driven by a degenerate white-in-time noise,
Journal of Differential Equations, 362, 230–249, 2023.
20. Chen, Peng; Shao, Qi-Man; **Xu, Lihu**
A probability approximation framework: Markov process approach,
Annals of Applied Probability, 33 (2), 1619–1659, 2023.
21. Deng, Changsong; Schilling, Rene; **Xu, Lihu**

- Singular integrals of subordinators with applications to structure properties of SPDEs*,
Transactions of the American Mathematical Society, 375, 6043-6073, 2022.
22. Chen, Peng; Lu, Jianya; **Xu, Lihu**
Approximation to stochastic variance reduced gradient Langevin dynamics by stochastic delay differential equations,
Applied Mathematics & Optimization, 85 no. 2, Paper number 15, 2022.
 23. Chen, Peng; Nourdin, Ivan; **Xu, Lihu**; Yang, Xiaochuan; Zhang, Rui
Non-integrable stable approximation by Stein's method,
Journal of Theoretical Probability, no. 2, 1137-1186, 2022.
 24. Lu, Jianya; Tan, Yuzhen; **Xu, Lihu**
Central limit theorem and self-normalized Cramer type moderate deviation for Euler-Maruyama scheme,
Bernoulli, 28 (2), 937-964, 2022.
 25. Chen, Peng; Jin, Xinghu; Li, Xiang; **Xu, Lihu**
A generalized Catoni's M-estimator under finite α -th moment assumption with $\alpha \in (1, 2)$,
Electronic Journal of Statistics, 15 (2), 5523-5544, 2021.
 26. Budhiraja, Amarjit; Chen, Yong; **Xu, Lihu**
Large deviations of the entropy production rate for a class of Gaussian processes,
Journal of Mathematical Physics, 62 (5), 052702, 2021.
 27. Shen, Hao; Song, Jian; Sun, Rongfeng; **Xu, Lihu**
Scaling limit of a directed polymer among a Poisson field of independent walks,
Journal of Functional Analysis, 281 (5), 109066, 2021.
 28. Sun, Xiaobin; Wang, Ran; **Xu, Lihu**; Yang, Xue
Large deviation for two-time-scale stochastic Burgers equation,
Stochastics and Dynamics, no. 5, Paper No. 2150023, 37 pp, 2021.
 29. Chen, Peng; Nourdin, Ivan; **Xu, Lihu**
Stein's method for asymmetric alpha-stable distribution, with applications to stable CLT,
Journal of Theoretical Probability, 34 (3), 1382-1407, 2021.
 30. Wang, Ran; Xiong, Jie; **Xu, Lihu**
Large deviation principle of occupation measures for non-linear monotone SPDEs,
Science China Mathematics, 64 (4), 799-822, 2021.
 31. Dong, Zhao; Wang, Feng-Yu; **Xu, Lihu**
Irreducibility and asymptotics of stochastic Burgers equation driven by alpha-stable processes,
Potential Analysis, 52, 371-392, 2020.
 32. Jia, Jinzhu; Xie, Fang; **Xu, Lihu**
Sparse Poisson regression with penalized weighted score function,
Electronic Journal of Statistics, 13(2), 2898-2920, 2019.
 33. Chen, Peng; **Xu, Lihu**
Approximation to stable law by the Lindeberg principle,
Journal of Mathematical Analysis and Applications, 480(2), 123338, 28 pp, 2019.
 34. Sun, Xiaobin; Xie, Yingchao; **Xu, Lihu**
Exponential mixing for SPDEs driven by highly degenerate Lévy noises,
Illinois Journal of Mathematics, 64(1), 75-102, 2019.
 35. **Xu, Lihu**
Approximation of stable law in Wasserstein-1 distance by Stein's method,
Annals of Applied Probability, 29(1), 458-504, 2019.
 - * This paper 'breaks new grounds on the famous Stein's method and opens up many new interesting research problems' (from the 5 pages' report of a referee).
 - * 'I'll look at this with real interest. You must have done something because a generation of probabalists have not been able to solve this. I'll give a read and report. Very glad that you made progress on this' (from Persi Diaconis' email).
 36. Fang, Xiao; Shao, Qi-Man; **Xu, Lihu**
Multivariate approximations in Wasserstein distance by Stein's method and Bismut's formula,
Probability Theory and Related Fields, 174(3,4), 945-979, 2019.
 37. Sun, Xiaobin; Xiao, Yimin; **Xu, Lihu**; Zhai, Jianliang
Uniform dimension results for a family of Markov processes,
Bernoulli, Vol. 24 4B, 3924-3951, 2018.
 38. **Xu, Lihu**
Singular integrals of stable subordinator,
Statistics & Probability Letters, Vol. 139, 115-118, 2018.
 39. Wang, Ran; **Xu, Lihu**
Asymptotics for stochastic reaction-diffusion equation driven by subordinate Brownian motion,
Stochastic Processes and Their Applications, Vol. 128 5, 1772-1796, 2018.

40. Xie, Fang; **Xu, Lihu**; Yang, Youcai
Lasso for sparse linear regression with exponentially beta-mixing errors,
Statistics & Probability Letters, Vol. 125, 64-70, 2017.
41. Wang, Ran; Xiong, Jie; **Xu, Lihu**
Irreducibility of stochastic real Ginzburg-Landau equation driven by alpha-stable noises and applications,
Bernoulli, Vol. 23 2, 1179-1201, 2017.
42. **Xu, Lihu**; Yue, Wen; Zhang, Tusheng
Smooth densities of the laws of perturbed diffusion processes,
Statistics & Probability Letters, Vol. 119, 55-62, 2016.
43. Chen, Xiaohong; Shao, Qi-Man; Wu, Wei Biao; **Xu, Lihu**
Self-normalized Cramer-type moderate deviations under dependence,
Annals of Statistics, Vol. 44 4, 1593-1617, 2016.
44. Chen, Yong; Ge, Hao; Xiong, Jie; **Xu, Lihu**;
The large deviation principle and steady-state fluctuation theorem for the entropy production rate of a stochastic process in magnetic fields,
Journal of Mathematical Physics, Vol. 57 7, 073302, 2016.
45. Wang, Feng-Yu; Xiong, Jie; **Xu, Lihu**
Asymptotics of sample entropy production rate for stochastic differential equations,
Journal of Statistical Physics, Vol. 163 5, 1211-1234, 2016.
46. Dong, Zhao; Peszat, Szymon; **Xu, Lihu**
On some smoothening effects of the transition semigroup of a Lévy process,
Journal of Mathematical Analysis and Applications, Vol. 434 2, 1566-1580, 2016.
47. Winter, Matthias; **Xu, Lihu**; Zhai, Jianliang; Zhang, Tusheng
The dynamics of the stochastic shadow Gierer-Meinhardt system,
Journal of Differential Equations, Vol. 260 1, 84-114, 2016.
48. Wang, Feng-Yu; **Xu, Lihu**; Zhang, Xicheng
Gradient estimates for SDEs driven by multiplicative Lévy noise,
Journal of Functional Analysis, Vol. 269 10, 3195-3219, 2015.
49. Li, Fang; **Xu, Lihu**
Finite time blowup of the stochastic shadow Gierer-Meinhardt System,
Electronic Communications in Probability, Vol. 20, 1-13, 2015.
50. Wang, Ran; **Xu, Lihu**
Asymptotics of the entropy production rate for d-dimensional Ornstein-Uhlenbeck processes,
Journal of Statistical Physics, Vol. 160 5, 1336-1353, 2015.
51. **Xu, Lihu**
Exponential mixing of 2D SDEs forced by degenerate Lévy noises,
Journal of Evolution Equations, Vol. 14 2, 249-272, 2014.
52. Dong, Zhao; **Xu, Lihu**; Zhang, Xicheng
Exponential ergodicity of stochastic Burgers equations driven by α -stable processes,
Journal of Statistical Physics, Vol. 154 4, 929-949, 2014.
53. Wang, Feng-Yu; **Xu, Lihu**
Log-Harnack inequality for Grushin type semigroups,
Revista Matemática Iberoamericana, Vol. 30 2, 405-418, 2014.
54. **Xu, Lihu**
Ergodicity of the stochastic real Ginzburg-Landau equation driven by alpha-stable noises,
Stochastic Processes and Their Applications, Vol. 123 10, 3710-3736, 2013.
55. Alberverio, Sergio; Debussche, Arnaud; **Xu, Lihu**
Exponential mixing of the 3D stochastic Navier-Stokes equations driven by mildly degenerate noises,
Applied Mathematics & Optimization, Vol. 66 2, 273-308, 2012.
56. Wang, Feng-Yu; **Xu, Lihu**
Derivative formula and applications for hyperdissipative stochastic Navier-Stokes/Burgers equations,
Infinite Dimensional Analysis Quantum Probability and Related Topics, Vol. 15 3, 1250020, 2012.
57. Priola, Enrico; Shirikyan, Armen; **Xu, Lihu**; Zabczyk, Jerzy
Exponential ergodicity and regularity for equations with Lévy noise,
Stochastic Processes and Their Applications, Vol. 122 1, 106-133, 2012.
58. **Xu, Lihu**
A modified log-Harnack inequality and asymptotically strong Feller property,
Journal of Evolution Equations, Vol. 11 4, 925-942, 2011.
59. Wang, Feng-Yu; Wu, Jiang-Lun; **Xu, Lihu**
Log-Harnack inequality for stochastic Burgers equations and applications,
Journal of Mathematical Analysis and Applications, Vol. 384 1 SI, 151-159, 2011.

60. Dong, Zhao; **Xu, Lihu**; Zhang, Xicheng
Invariant measures of stochastic 2D Navier-Stokes equations driven by α -stable processes,
Electronic Communications in Probability, Vol. 16, 678-688, 2011.
61. Priola, Enrico; **Xu, Lihu**; Zabczyk, Jerzy
Exponential mixing for some SPDEs with Lévy noise,
Stochastics and Dynamics, Vol. 11 2-3 SI, 521-534, 2011.
62. Romito, Marco; **Xu, Lihu**
Ergodicity of the 3D stochastic Navier-Stokes equations driven by mildly degenerate noise,
Stochastic Processes and Their Applications, Vol. 121 4, 673-700, 2011.
63. **Xu, Lihu**; Zegarlinski, Boguslaw
Existence and exponential mixing of infinite white alpha-stable Systems with unbounded interactions,
Electronic Journal of Probability, Vol. 15, 1994-2018, 2010.
64. **Xu, Lihu**; Zegarlinski, Boguslaw
Ergodicity of the finite and infinite dimensional α -stable systems,
Stochastic Analysis and Applications, Vol. 27 4, 797-824, 2009.
65. Olkiewicz, Robert; **Xu, Lihu**; Zegarlinski, Boguslaw
Nonlinear problems in infinite interacting particle systems,
Infinite Dimensional Analysis Quantum Probability and Related Topics, Vol. 11 2, 179-211, 2008.

Teaching

- **Teaching at Michigan State University**
 1. STT 441: Probability (undergraduate)
- **Teaching at University of Macau**
 1. MATH 3025: Topics in Statistics and Data Science: Reinforcement Learning (undergraduate)
 2. MATH 3025: Topics in Statistics and Data Science: Bayesian Inference and MCMC (undergraduate)
 3. MATH 2010: Computational Statistics (undergraduate)
 4. MATH 3017: Data Driven Sampling Methods (undergraduate)
 5. MATH 2006: Applied Statistics (undergraduate)
 6. MATH 3008 (MATB 352): Introduction to Stochastic Calculus (undergraduate)
 7. MATH 2005 (MATB 213): Probability (undergraduate)
 8. MATH 325: Introduction to Real Analysis and Hilbert Spaces (undergraduate)
 9. CISB 122/ECEB 252/MATH 111: Probability and Statistics for Engineering (undergraduate)
 10. MATH 7011/MATH 8001: Topics in Probability and Statistics: High Dimensional Probability with Applications in Data Science (postgraduate)
 11. MATH 7013 (MATH 713): Stochastic Differential Equations (postgraduate)
 12. MATB 720: Advanced Probability (postgraduate)
 13. MATH 4004: Final Year Projects
 - Theory and applications of diffusion models (excellent project), 2025
 - Generative AI: diffusion models (excellent project), 2023
 - Tensor data analysis and its applications, 2022
 - A test of machine learning method by MNIST dataset (excellent project), 2021
 - Prediction of occurrence of influenza A and B in Macao using methodological factors, 2019
 - Central limit theorem for high dimensional regressions, 2018
 - Fused Kolmogorov filter: a nonparametric model-free screening method (excellent project), 2017
 - Dictionary identification-sparse matrix-factorization via ℓ_1 minimization, 2017
 - False discovery rate in statistics (excellent project), 2016
 - Least angle regression and square-root Lasso (excellent project), 2015
 14. MATH 7012: Master Reading Courses
 - Distributional Reinforcement Learning, 2025
 - Thompson Samplings, 2024
 - The Applications of Malliavin Calculus in Numerical Analysis, 2023
 - Random Tensor Data Analysis, 2022
 - Reinforcement Learning, 2021
 - Stochastic Gradient Descent, 2020
 - High Dimensional Statistics, 2019
 - Stein's methods, 2018
 - Lasso, 2017
 - Large Deviations, 2016

- **Teaching at Brunel University London and Technische Universität Berlin**
 1. Probability Theory (undergraduate), BU London
 2. Mathematical Analysis II (undergraduate), BU London
 3. Mathematical Analysis I (undergraduate), BU London
 4. Final year project course: Financial Mathematics (undergraduate), BU London
 5. Stochastic Partial Differential Equations (PhD), TU Berlin
- **Mini Courses**
 1. 'Approximation of stable law in Wasserstein-1 distance by Stein's method', Simon Semester, Banach International Mathematical Research Center, Warsaw, 07-08/06/2018.
<https://www.impan.pl/en/activities/banach-center/conferences/18simons-viii/program>
 2. 'Approximation of stable law in Wasserstein-1 distance by Stein's method', Wuhan University, Wuhan, 02-03/04/2018.
http://maths.whu.edu.cn/kexueyanjiu/xueshubaogao/index_5.html
- **Public Lectures on Open Days**
 1. 'Nobel Prize and Mathematics' (in Chinese), Open Day for High School Students, University of Macau, 01/2023
 2. 'A look at Nash equilibrium from McDonald and KFC' (in Chinese), Open Day for High School Students, 04/2018

Research Grants and Honors

- **Grants from the government of P.R. China**
 1. National Natural Science Foundation of China, NSF 12071499 (**general project**)
Period: 01/01/2021-31/12/2024
Title: *Approximation to ergodic distributions of stochastic (partial) differential equations and applications*
Amount: RMB 510,000
Role: **PI**
 2. National Natural Science Foundation of China, NSF 11571390 (**general project**)
Period: 01/01/2016-31/12/2019
Title: *Ergodicity of Stochastic PDEs driven by highly degenerate pure jump noises*
Amount: RMB 520,000
Role: **PI**
- **Grants from the government of SAR Macau**
 1. The Science and Technology Development Fund, FDCT/0074/2023/R1A2
Period: 01/01/2024-31/12/2026
Title: *Applications of methods in stochastic analysis to stochastic algorithms in data science*
Amount: MOP 1,264,000
Role: **PI**
 2. The Science and Technology Development Fund, FDCT/0090/2019/A2
Period: 01/01/2020-31/12/2022
Title: *Applications of Stein's method in high dimensional probability approximations*
Amount: MOP 1,237,000
Role: **PI**
 3. The Science and Technology Development Fund, FDCT 038/2017/A1
Period: 01/09/2017-30/09/2020
Title: *Fluctuation of entropy production rate*
Amount: MOP 565,000
Role: **PI**
 4. The Science and Technology Development Fund, FDCT/025/2016/A1
Period: 01/12/2016-30/11/2019
Title: *Some optimization problems with model ambiguity and information incompleteness*
Amount: MOP 1,463,000
Role: **PI**
 5. The Science and Technology Development Fund, FDCT/030/2016/A1
Period: 01/09/2016-31/08/2019
Title: *Large and moderate deviations for empirical measure of stochastic partial differential equations*
Amount: MOP 850,000
Role: **PI**
 6. The Science and Technology Development Fund, FDCT/049/2014/A1
Period: 20/01/2015-19/01/2017
Title: *Stochastic Gierer-Meinhardt and Schnakenberg models in biology*
Amount: MOP 535,000
Role: **PI**

- **Grants from University of Macau**

1. Multi-Year Research Grant, MYRG-GRG2023-00088-FST
Period: 01/01/2024-31/12/2025
Title: *Distribution estimations of heavy tailed time series by DNN-based GANs*
Amount: MOP 150,000
Role: **PI**
2. Multi-Year Research Grant, MYRG2020-00039-FST
Period: 01/01/2022-31/12/2023
Title: *Applications of Stein's method in queueing models*
Amount: MOP 175,000
Role: **PI**
3. Multi-Year Research Grant, MYRG2018-00133-FST
Period: 01/01/2019-31/12/2021
Title: *High dimensional probability approximations*
Amount: MOP 545,000
Role: **PI**
4. Multi-Year Research Grant, MYRG2016-00025-FST
Period: 01/01/2017-31/12/2019
Title: *Applications of coupling method to studying the ergodicity of stochastic PDEs driven by degenerated pure jump Lévy processes*
Amount: MOP 1,110,000
Role: **PI**
5. Matching Grant, MRG017/XLH/2015/FST
Period: 01/08/2014-31/07/2015
Title: *Stochastic Gierer-Meinhardt and Schnakenberg models in biology*
Amount: MOP 178,000
Role: **PI**
6. Multi-Year Research Grant, MYRG2016-00025-FST
Period: 01/04/2015-31/03/2018
Title: *Applications of the LASSO and the Square-root LASSO methods to correlated big data*
Amount: MOP 672,000
Role: **PI**
7. Starting-Up Research Grant, SRG2013-00064-FST
Period: 01/01/2014-31/01/2015
Title: *Large deviation of stochastic Gierer-Meinhardt and Schnakenberg models*
Amount: MOP 100,000
Role: **PI**

- **Honors**

- Teaching Excellence Award of Faculty of Science and Technology
University of Macau, 2023/2024
- Teaching Excellence Award of Faculty of Science and Technology
University of Macau, 2018/2019
- Research Excellence Award of Faculty of Science and Technology
University of Macau, 2016/2017
- Invited Lecturer of Mini Course
Simon Semester, Banach International Mathematical Center, Warsaw, Poland, 2018
- Junior Scientific Leader
Banach International Mathematical Center, Warsaw, Poland, 2018
- DAAD Fellowship
Berlin, Germany, 2006
- Overseas Research Student (ORS) Award
The government of UK, 2004–2007
- International Student Scholarship
Imperial College London, UK, 2004–2007

Supervised Students and Mentored Postdocs

- Postdoc (1)
 - (a) **Dr. Huiming ZHANG** (2020-2022)
First Institute: Tenure Track Associate Professor at Beihang University (Beijing University of Aeronautics and Astronautics)
- Graduated Supervised PhD Students (9)
 - (a) **Dr. Fang XIE** (2015-2018)
Thesis: ℓ_1 -penalized method in high dimensional regressions and its theoretical properties

First Job: Postdoc at Ruhr-Universitaet Bochum
Current Job: Assistant Professor at UIC of BNU-HKBU

- (b) **Dr. Xinghu JIN** (2017-2021)
Thesis: *Approximation to ergodic measure of queueing systems*
First Job: Assistant Professor at Hefei University of Technology
- (c) **Dr. Peng CHEN** (2017-2021)
Thesis: *Probability approximations with their applications to limit theorems and stochastic algorithms*
First Job: Assistant professor at Nanjing University of Aeronautics and Astronautics
Current Job: Associate professor at Nanjing University of Aeronautics and Astronautics
- (d) **Dr. Xin XU** (2017-2021)
Thesis: *Aggregation in several chemotaxis models* (joint with Xuefeng Wang and Linlin Su)
First Job: Postdoc at Nankai University
Current Job: Assistant Professor at South China Normal University
- (e) **Dr. Jianya LU** (2018-2022)
Thesis: *Limit theorems for high dimensional time series and stochastic algorithms*
First Job: Lecturer in Statistics at the University of Essex, UK
[See UM news.](#)
- (f) **Dr. Qiuran YAO** (2019–2023)
Thesis: *Robust estimations in statistical learning and distribution estimations via DNN-based GAN* (joint with Lianjie Shu)
First Job: Assistant Professor at Zhuhai City Polytechnic
- (g) **Dr. Xiang LI** (2020–2024)
Thesis: *Non-asymptotic convergence analysis of stochastic processes and related algorithms*
First Job: Postdoc at Southern University of Science and Technology
- (h) **Dr. Yu WANG** (2021–2025)
Thesis: *Unified analysis of stochastic differential equations: Euler-Maruyama scheme, underdamping Langevin dynamics, and coupling methods with Lévy noise*
First Job: Assistant Professor at Yunnan Normal University
- (i) **Dr. Yingjun MO** (2022–2026)
Thesis: *Stochastic differential equations with non-globally Lipschitz drift: limit theorems and tamed Euler-Maruyama schemes*
First Job: Postdoc at Southern University of Science and Technology
- Graduated Co-supervised PhD Students (1)
 - (a) **Dr. Hao XU** (2021–2025)
Thesis: *Hessian spectrum at the global minimum, topology trivialization and expected number of critical points of locally isotropic Gaussian random fields*
First Job: Assistant Professor at Central South University
- Ongoing PhD Students (3)
 - (a) **Mr. Qiyang PEI** (2023–)
Research Topic: *Quantitative central limit theorem and approximation related to queues with heavy tails*
 - (b) **Mr. Zhihao ZHANG** (2024–)
Research Topic: *Stochastic analysis for MABs and RL.*
 - (c) **Mr. Bo WU** (2025–)
- Graduated Master's Students (10)
 - (a) **Mr. Libang DENG** (2013-2015)
Thesis: *Asset price model and an SPDE*
First Job: Huashi Finance Co. Ltd
 - (b) **Ms. Fang XIE** (2013-2015)
Thesis: *Square-root Lasso for mixing time series*
First Job: PhD at UM
 - (c) **Mr. Zhilei FAN** (2015-2017)
Thesis: *False discovery rate and self-normalized Cramer type moderate deviations*
First Job: Beijing Tao Capital Company Limited
 - (d) **Ms. Qiuran YAO** (2016-2019)
Thesis: *Lasso and estimation consistency of Tweedie's compound Poisson model with elastic net*
First Job: PhD at UM
 - (e) **Ms. Jingjing YU** (2018-2021)
Thesis: *Representer Theorems in Reproducing Kernel Hilbert Space*
First Job: PhD at UM
 - (f) **Mr. Xiang LI** (2018-2020)
Thesis: *Catoni's estimator for heavy tailed data and stable approximations*
First Job: PhD at UM
 - (g) **Mr. Man-Nok LOU** (2019-2021)

Thesis: *Applications of linear regressions*

First Job: Pei Tao Secondary School

(h) **Mr. Qiyang PEI** (2021–2023)

Thesis: *Malliavin calculus and its applications in stochastic numerical scheme*

First Job: PhD at UM

(i) **Mr. Yueren ZHAO** (2023–2025)

Thesis: *Thompson samplings and their applications*

First Job: Zhengzhou School of Finance

(j) **Mr. Tiansui TU** (2025–2026)

Thesis: *Posterior analysis of manifold-constrained Gaussian process inference for simple ordinary differential equation models*

First Job: PhD at Chinese University of Hong Kong (won the prestigious [Hong Kong PhD Fellowship Scheme](#))

• Undergraduate Students in My Research Group (**16**)

(a) **Mr. Yong MA** (2015): Master at UM & Ritsumei University

(b) **Mr. Zhiyuan YU** (2016): PhD at Florida State University

(c) **Mr. Jingxiu WANG** (2017): Guangzhou Netease Company

(d) **Mr. Mu LI** (2017): Master at National University of Australia

(e) **Mr. Bo WU** (2018): Master at Wake Forest University with full studentship

(f) **Ms. Zhe LI** (2019): Master at University of Edinburgh

(g) **Ms. Shengqi CHENG** (2021): Master at University of Sun Yat Sen

(h) **Ms. Jialin ZENG** (2021): Master at UM

(i) **Ms. Houpu GAO** (2022): Master at Columbia University

(j) **Ms. Yuan TIAN** (2022): Master at University of Edinburgh

(k) **Ms. Sinuo LIU** (2023): Master at ETH Zurich

(l) **Ms. Yunjin ZHANG** (2023): Master at University of Wisconsin-Madison

(m) **Mr. Sihan LONG** (2025): Master at Brown University

(n) **Mr. Tianyun ZHAO** (2025): Master at National University of Singapore

(o) **Mr. Yushi ZHAO** (2025): Master at City University of Hong Kong

• Honor project undergraduate students (**3**)

• Summer research students (**8**)

• Undergraduate students at colleges (**> 10**)

• Visiting PhD students (**> 10**)

Talks

• **Keynote or Colloquium Talks**

• 'Recent advances in stochastic approximations and their applications',
Colloquium Talk, Michigan State University, 02/2025.

• 'Catoni type robust estimation for heavy tailed data',
Colloquium Talk, Michigan State University, 02/2024.

• 'A framework for stochastic approximations with its applications',
7th Conference of Probability in China, Weihai, 09/2022.

<https://probability2022.casconf.cn>

• 'Approximation of stable law by Stein's method',
11th Congress of Probability and Statistics in China, Chengdu, 10/2018.

<http://stat.swufe.edu.cn/probstat2018>.

• 'Approximation of stable law by Stein's method',
10th Conference of Limit Theorem in China & 2nd Sino-Russian probability and large sample conference, Changchun, 09/2017.

<http://klas.nenu.edu.cn/info/1017/1175.htm>

• **Selected Invited Talks**

• Stein's method and its applications
Invited talk, Probability Webinar of University of Minnesota, 03/2026.

• Quantitative bounds for high dimensional entropic CLT
Invited talk, IMS Asia Pacific Rim Meeting, 06/2026.

• Stein's method and related approximation
Invited talk, The third biennial conference HKSIAM, 07/2025.

• Stein's method and related approximation
Invited talk, The 10th Symposium on BSDE, 06/2025.

• Jointly adaptive robust estimations
Invited talk, ICSA China Conference, Zhuhai, 06/2025.

• Stable central limit theorem in total variation distance

- International Workshop on Stochastic Analysis, University of Science and Technology of China, 03/2025
- Phase transition in an EM scheme for SDEs driven by stable process
International Workshop on Probability Theory, Wuhan University, 08/2024.
 - Robust heavy tailed estimations
University of Tennessee, Knoxville, 04/2024.
 - Approximation to invariant measure of SDE driven by stable distribution
The 2nd HKSIAM Biennial Conference (stochastic dynamics session), Hong Kong, 09/2023.
 - Bounds of Langevin algorithms in the Wasserstein- p distance with $p \in [0, 1]$
International Conference on Markov Processes, Tianjin, China, 07/2023
 - A probability approximation framework with applications
Joint Conference on Statistics and Data Science in China (IMS), Peking University, 07/2023.
 - Robust high dimensional statistical learning on heavy tailed data,
The 12th ICSA International Conference, Hong Kong, 07/2023.
 - A probability approximation framework via Markov process approach,
Hong Kong/Singapore joint Seminar in Financial Math/Engineering, 14/08/2022, Zoom.
 - Approximation to ergodic measure of stable SDEs by EM scheme,
5th International Conference on Econometrics and Statistics (Stein's method session), Kyoto, 06/2022.
 - A general probability approximation framework with its applications,
Nanjing University, Nanjing, 06/2022.
 - Approximation to SVRG algorithms by stochastic delay equations,
Peking University, Beijing, 04/2022.
 - Self-normalized moderate deviation of Euler-Maruyama scheme,
Southern China University of Technology, 29/04/2021.
 - Approximation of stable law,
Ritsumeikan University Seminar, Shiga (Japan), 05/2021.
 - Approximation of stable law,
World Congress on Statistics (Stein's method session), Kuala Lumpur, 08/2019.
 - Approximation of probability distributions,
European Mathematical Society & Bernoulli Society Meeting (Stein's method session), Palermo, 07/2019.
 - Member of a research collaboration workshop 'Stein's method and high dimensional Statistics',
American Institute of Mathematics, San Jose, California, 08/2018
<https://aimath.org/workshops/upcoming/steinhd/>
 - Talk 1: A Malliavin-Stein approach for multivariate approximations in Wasserstein distance,
The 12th AIMS Conference on Dynamical Systems, Differential Equations and Applications, Taipei, 07/2018
 - Talk 2: Approximation of stable law in Wasserstein-1 distance by Stein's method,
The 12th AIMS Conference on Dynamical Systems, Differential Equations and Applications, Taipei, 07/2018
 - A Malliavin-Stein approach for multivariate approximations in Wasserstein distance,
Conference of Stochastic Processes and Their Applications, Gothenburg, 06/2018
 - Approximation of stable law under Wasserstein-1 distance,
Hong Kong Joint Probability Seminar, Hong Kong, 05/2018.
 - Approximation of stable law,
International conference on probability theory and statistical physics, Beijing, 11/2017.
 - Convergence rate of stable law,
CNRS-IMPAN mathematics summer institute conference, Krakow, 07/2017.
 - Convergence rate of stable law,
International conference of Markov processes and their related fields, Wuhan, 07/2017.
 - The long time behaviour of stochastic system with singular coefficients,
10th ICSA international conference, Shanghai, 12/2016.
 - The Large Deviation Principle and Steady-state Fluctuation Theorem for the Entropy Production Rate of a Stochastic Process in Magnetic Fields,
Workshop of Young Probabilists, Hefei, 08/2016.
 - The Large Deviation Principle and Steady-state Fluctuation Theorem for the Entropy Production Rate of a Stochastic Process in Magnetic Fields,
Conference of Stochastic Dynamics and SPDEs, Shanghai, 07/2016.
 - Large deviation for monotone SPDEs,
The 4th Institute of Mathematical Statistics Asia Pacific Rim Meeting, Hong Kong, 06/2016.
 - Large deviation for monotone SPDEs,
8th International Conference on Stochastic Analysis and Applications, Beijing, 06/2016.
 - Asymptotics of the entropy production rate for d -dimensional Ornstein-Uhlenbeck processes,
2015 IMS-China International Conference on Statistics and Probability, Kunming, 07/2015.
 - Asymptotics of the entropy production rate for d -dimensional Ornstein-Uhlenbeck processes,
Workshop of Young Probabilists at Peking University, Beijing, 07/2015.

- Asymptotics of the entropy production rate for d -dimensional Ornstein-Uhlenbeck processes, Workshop of Young Probabilists, Wuhan, 05/2015.
- Ergodicity of SPDEs driven by degenerate α -stable noises by coupling method, Workshop of Stochastic Analysis and its Applications, Xuzhou, 12/2014.
- Ergodicity of Stochastic PDEs driven by α -stable noises, Workshop of Infinite Dimensional Analysis and SPDEs, London, 05/2014.

Services

- **Editorial Work**
 - Associate Editor of Journal of Theoretical Probability, 2026–now
- **Services to University/Faculty/Department**
 - Member of PhD Admission Committee of Department of Mathematics, 2018–now
 - Fellow of Chao Kuang Piu College, 2015–now
 - Member of Principal Recommendation Students Admission Committee, 2015–now
 - Member of Master Students Admission Committee, 2015–now
 - Member of Committee of Final Year Project of Department of Mathematics, 2015–now
 - Member of Faculty Association of University of Macau, 2014–now
 - Marker of Joint Admission Examination of Macau Higher Education Institutions, 2016–2018
 - Good Suggestions Award of University of Macau, 2015
 - Member of New Academic Staff Club, 2014
 - Service to the development of the undergraduates and graduate students
 - * Supervisor of Undergraduates' Summer Research, 2017, 2020, 2021, 2022
 - * Mentor of Undergraduates at Chao Kuang Piu College, 2017–now
 - * 'Tea With Professors' Activity, 2014 & 2017
 - * Recommendation letters for undergraduates and postgraduates > **60**
 - * Member of PhD Advisory Committees > **20**
 - * Assessment Panel of PhD Qualifying Examinations > **20**
- **Services with Leadership**
 - Chair of the committee 'Undergraduate Summer Camp', DoM, UM, 2018–2024
 - Chair of 'Minor Program in Mathematics', UM, 2016–now
 - PI of 'Mathematics of Machine Learning Section' of the Center for Applied Mathematics, UM, 2020–now
 - Leader of the research group 'Applied Probability', DoM, UM, 2018–now
 - Leader of Undergraduates' Academic Visit to Sun Yat-Sen University, DoM, UM, 2017
- **Professional Services**
 - **External referees**
 - * Grant application from Czech Science Foundation, 2024
 - * Grant applications from ERC, 2021
 - * Grant applications from EPSRC of UK, 2018
 - * Grant applications from NSF of China, 2016–now
 - * Grant applications from TianYuan Foundation of Mathematics, 2015
 - **Referees for journals: > 60 papers**
 Journal of American Mathematical Society (2 times), Annals of Statistics, Annals of Probability, Journal of Differential Equations, Annals of Applied Probability, Bernoulli, Journal of Illinois Mathematics, Stochastic Processes and Their Applications, Applied Mathematics & Optimization, Statistics & Probability Letters, Sciences in China-Mathematics, Potential Analysis, Communications in Pure and Applied Analysis, Mathematische Nachrichten, Stochastic Analysis and Its Applications, Journal of Computational and Applied Mathematics, Journal of Applied Probability, Journal of Theoretical Probability, Electronic Communications in Probability, Journal of Mathematical Analysis and its Applications, East Asian Journal on Applied Mathematics, Econometrics Review.
 - **Organizers**
 - * 19th International Symposium on Econometric Theory and Applications (local organizer committee), 06/2025.
 - * The symposium 'Interplay of stochastic dynamics and machine learning', Conference EASIAM 2024, Macau, 07/2024.
 - * The session 'Nonnormal approximations and their applications', The 6th Institute of Mathematical Statistics Asia Pacific Rim Meeting, Melbourne, 01/2024.
 - * The seminar 'Probability and Statistics', University of Macau, 2016–2020 & 2022–now.
 - * The webinar 'Probability and Statistics', University of Macau, 2020–2022.
 - * The interdisciplinary seminar 'Machine Learning and its Applications' (Joint with Prof. Weijia JIA), University of Macau, 2019–2021.
 - * The workshop 'Stein's method and its related topics', University of Macau, 2018
 - * The session 'Statistical Physics' at 'Joint Annual Conference of Physics Societies in Greater Bay Area', 2018
 - * The workshop 'Probability Theory and its Applications', University of Macau, 2015

* The session 'Recent Advances in Stochastic Processes and its Applications', World Congress of Statistics, 2013

- Reviewer of American Mathematical Society, 2011-now
- Member of Macau Bioinformatics Association Committee, 2018-now
- Junior Scientific Leader of Banach International Mathematical Center, 2018
- Junior Member of Financial Mathematics Committee of SIAM China, 2017-now
- Memberships
 - Committee Member of Junior Financial Mathematics Society of China
 - Committee Member of Physics Society of Macau
 - Member of American Mathematical Society
 - Member of Macau Bioinformatics Association
 - Member of SIAM China
- **Community-Related Services**
 - Public Lecture 'Nobel Prize and Mathematics', Open Day for High School Students, University of Macau, 01/2023
 - High School's Public Lecture 'Suggestions for how to study mathematics and the daily examples of mathematics', Hao Jiang High School, 11/2022
 - Public Lecture 'Tips for studying in America and Europe', Cheng Yu Tung Residential College, University of Macau, 04/2022
 - Public Lecture 'A look at Nash equilibrium from McDonald and KFC', Open Day for High School Students , 04/2018
 - Public Lecture 'Probability in our daily life' at Chao Kuang Piu Residential College, University of Macau, 11/2017
 - Full Marathon in Macau (4 hours 44 mins), Macau, 12/2014
<https://www.macaomarathon.com/pt/results?gender=M&year=2014>

Skills

- Languages: English, Chinese
- Computer Skills: C, Matlab, Python.